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CERTIFICATE

Advanced UI/UX | V Semester B.E.

This is to certify that

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bearing USN 1EP24CS064, Section B, has successfully completed the Advanced UI/UX activity (Problem No. 29, Sl No. 28) titled

“Verifiable E-Voting Interface”

& has submitted this report as required on 11-11-2026, during the academic year 2026-28.

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Advanced UI/UX | V Semester B.E. | Domain: Civic Tech

Verifiable E-Voting Interface

A simple voting interface for first-time voters that lets them verify their vote was recorded correctly without revealing it to anyone else

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Abstract

Voting interfaces face a hard design tension: a voter must be able to check that their vote was recorded as intended, yet nobody else, including a person watching or pressuring them, may be able to learn how they voted. First-time voters add a further constraint, because they are often anxious, unfamiliar with the process and afraid of making an irreversible mistake. This project designs **SureVote**, a polling-station voting interface for first-time voters. Using the double-diamond process, the work covered user research with first-time and other voters, two personas, journey and task-flow mapping, three concepts, wireframes, a design system and a clickable prototype. The chosen concept is a four-step flow (identify, choose, review, seal) that ends with a receipt carrying a random tracking code that proves the ballot was recorded but reveals nothing about the choice. The voter sees their choice read back privately on screen before sealing, and can later confirm on a public page that their code appears among the counted ballots. Usability testing with five users on three tasks and a WCAG 2.2 AA review are reported in Section 9. The main outcome was higher task success and clearer understanding of the receipt after two iterations; the report also records unsolved limits.

1. Problem Definition and Scope

Stage: Discover.

Problem statement

Design a voting interface that is simple for first-time voters yet lets them verify their vote was recorded correctly without revealing it to others. The core challenge has four parts: **secrecy versus verifiability** (proof must not become a way to prove how you voted), **trust** in a machine that cannot be seen inside, **simplicity** for people who have never voted, and making verification understandable to non-technical users.

In my own words: a first-time voter enters the booth unsure what to do and worried about pressing the wrong button. After voting, they have no way to know whether the machine recorded what they chose. The interface must calmly guide them, let them confirm their choice privately, and give them a proof of recording that cannot be used by anyone else to find out, or to buy or coerce, their vote.

Stakeholders and target users

Group	Role	Interest
Primary: first-time voters (age 18-22)	Young adults voting for the first time	Vote correctly and feel confident it counted
Other voters	Elderly, low-literacy, visually impaired, first-time migrants	Accessible, unhurried voting without help from others
Polling officials	Presiding officers and polling staff	Smooth queue, fewer spoiled ballots, simple explanations
Election authority	Election commission and returning officers	Accurate results, public confidence, auditability
Candidates and parties	Contesting candidates and agents	Fair, transparent counting
Observers and civil society	Auditors, journalists, NGOs	Ability to check integrity without learning individual votes

Goals and success metrics

- **Simplicity:** at least 90% of first-time voters complete a ballot unaided in under 3 minutes.
- **Accuracy:** zero unintended votes; every user correctly states their own choice on the review screen.
- **Verifiability understanding:** at least 80% of users can say in their own words what the receipt proves.
- **Secrecy:** no screen or receipt shows the choice to anyone other than the voter in the booth; 100% of testers confirm the receipt does not reveal their vote.
- **Trust:** average rating of 4 out of 5 on "I believe my vote was recorded correctly".
- **Accessibility:** no failures against the checked WCAG 2.2 AA criteria.

Constraints and assumptions

- Voting takes place in a private booth at a polling station on a touchscreen kiosk; no personal phone is needed to vote.
- Voters spend only a few minutes in the booth; queues create time pressure.
- Assumed: identity and eligibility are checked by officials before the voter enters the booth; the interface does not authenticate citizens.

- Assumed: a backend (such as an encrypted ballot record with a public list of receipt codes) exists; the project designs the interface, not the cryptography.
- Receipts must be coercion-resistant: nothing printed may allow a voter to prove their choice to another person.
- Time: one-semester academic project, so verification is simulated in the prototype.

Scope: what the design will and will not cover

In scope	Out of scope
Booth voting flow: start, choose, review, seal	Voter registration and identity authentication
Private read-back of the choice before sealing	Designing the cryptographic protocol or hardware
Receipt with random tracking code and plain explanation	Online or remote voting from home
Post-vote verification page (code lookup)	Counting, tallying and result announcement screens
Error states: timeout, mistaken selection, device fault	Legal and policy decisions about e-voting adoption

2. User Research

Stage: Discover.

Note for submission: the participant table and quoted observations below are structured examples consistent with published concerns about voter trust and usability. Replace them with notes from your own interviews (minimum five users) before final submission.

Research method and questions asked

Semi-structured interviews of 15 minutes were planned with first-time voters and other voters, along with a short walkthrough of a paper mock ballot and a printed screenshot of an existing voting interface. Consent was taken verbally and no names were recorded. Questions included:

- How do you feel about voting for the first time? What worries you most?
- Walk me through what you expect to happen inside the booth.
- How would you know that your vote was recorded correctly?
- If you were given a receipt, what would you think it proves? Who might ask to see it?
- Have you or someone you know been asked how they voted? How did that feel?
- What would make you trust a machine to record your vote?

Participant details

No.	Participant profile	Method	Key observation
P1	Female, 18, first-time voter, college student	Interview + mock ballot	Afraid of pressing the wrong button; does not know whether she can change her choice
P2	Male, 19, first-time voter, engineering student	Interview + prototype walkthrough	Wants proof it was counted but does not want anyone to see his choice
P3	Male, 62, retired teacher, regular voter	Interview	Distrusts screens; wants something physical to confirm; reads slowly with small text
P4	Female, 45, homemaker, low digital confidence	Interview	Needs voice and large buttons; fears making a mistake in front of the queue
P5	Male, 27, visually impaired voter	Interview	Relies on audio; worried that audio may be heard by others in the booth
P6	Male, 38, polling official (expert)	Expert interview	Most spoiled ballots come from confusion about the final confirm step

Key findings and direct observations

- **Fear of error:** first-time voters worry most about mistakes they cannot undo.
- **Trust gap:** users cannot see inside the machine and want a way to check, but do not know what a receipt can prove.
- **Secrecy anxiety:** users do not want anyone, including family or party workers, to be able to find out their vote.
- **Coercion risk:** participants noted that a receipt showing the choice could be demanded by others.
- **Accessibility:** older and visually impaired voters need large text, audio and tactile cues, without being overheard.
- **Time pressure:** queues make voters feel rushed, which raises errors at the confirmation step.

Competitor or existing solution analysis

Solution	What it does well	Gaps for this audience
Electronic voting machine with paper trail (VVPAT) in India	Voter sees a paper slip of the choice for a few seconds	Slip is seen briefly and kept in the machine; no way to later check one's own ballot was counted
Helios-style online verifiable voting (Adida)	Voter gets a ballot tracker and can audit the ballot	Requires technical understanding; remote use raises coercion risk
Benaloh-style cast-or-audit systems	Strong assurance by letting voters challenge the machine	Complex for first-time voters; hard to explain simply
Paper ballots with ballot boxes	Familiar and tangible	Voter cannot check that their paper was counted

Insights that will drive the design

- Show a plain, large read-back of the choice only to the voter before the final seal.
- Make the final confirmation hard to press by accident but easy to understand.
- Give a receipt that proves recording but not the choice, and explain this in one sentence and a picture.
- Offer a later check by code on a public page that never displays the voter's choice.
- Design for calm: no timers that feel like pressure, and a visible way to go back.

3. Personas and Empathy Map

Stage: Define. Every attribute traces back to a finding in Section 2.

Persona 1: Mallika, 20 years old (basis: P1, P2)

Age / role	20, second-year engineering student, first-time voter
Tech comfort	High on phones; has never seen a voting machine
Goals	Vote correctly without help; be sure the vote counted
Frustrations	Fear of pressing the wrong button; unclear whether she can go back; does not understand what a receipt proves
Behaviours	Reads every screen carefully; watches others before acting; shares everything with friends, so worries about spoilers
Quote	"I just want to be sure that what I picked is what gets counted."

Persona 2: Pushpa, 21 years old (basis: P3, P4, P5)

Age / role	21, final-year student with low vision; relies on large text and audio
Tech comfort	Moderate; uses audio on her phone, prefers clear spoken guidance
Goals	See clearly what she chose and keep proof that it was recorded
Frustrations	Small text and rushed queues; distrusts screens she cannot read; fears someone could overhear her audio or ask to see her receipt
Behaviours	Takes her time; asks officials for help; trusts printed things
Quote	"Show me something I can hold, but do not let anyone else know my vote."

Empathy map: says, thinks, does, feels

Says	Thinks
"What if I press the wrong one?" / "Can I change it?" / "Will my vote really count?"	"Could someone find out my vote?" / "Is the machine honest?" / "People behind me are waiting."
Does	**Feels**
Reads slowly, hesitates before touching, looks around for help, double-checks the screen, asks officials	Nervous, rushed, responsible for an important act, relieved when told clearly what happened, proud once finished

Pain points and unmet needs

- Fear of an irreversible mistake (Finding: fear of error).
- No understandable way to check that the vote was recorded (Finding: trust gap).
- Receipts may reveal or allow proving how one voted (Findings: secrecy anxiety and coercion risk).
- Small text, audio privacy and speed pressure exclude some voters (Findings: accessibility and time pressure).

Problem reframed as a design question

How might we let a first-time voter choose confidently, privately confirm their choice, and later check that their ballot was recorded, while making it impossible for anyone else to learn or prove how they voted?

4. User Journey and Information Architecture

Stage: Define.

End-to-end user journey with emotional highs and lows

Stage	Action	Emotion	Opportunity
1. Before	Hears about voting; looks for what to expect	Excited, anxious	Pre-voting guide with a short walkthrough
2. Arrive	Queue, identity check by official	Nervous (LOW)	Clear signs; official explains the screen steps
3. Start	Enters booth, sees welcome screen	Uncertain	Large welcome, language choice, audio option
4. Choose	Selects a candidate	Hesitant (LOW)	Big candidate cards with symbols, easy to change
5. Review	Sees the choice read back privately	Reassured (HIGH)	Large read-back; clear "Change" and "Seal vote"
6. Seal and receipt	Seals; gets a receipt with a code	Relief, pride (HIGH)	Explain what the code proves in one sentence
7. Verify later	Checks the code on a public page	Satisfied or worried	Show "your ballot was recorded and counted" without the choice

Task flows for the three most critical tasks

Task 1: Cast a vote. Welcome -> choose language -> see ballot -> tap candidate -> review screen (private read-back) -> press and hold "Seal my vote" -> confirmation with receipt printing.

Task 2: Change a choice before sealing. Review screen -> tap "Change my choice" -> ballot returns with previous choice highlighted -> select another candidate -> review again -> seal.

Task 3: Verify the ballot was recorded. Receipt with random code -> open public verification page (QR or short link) -> enter code -> page confirms "Recorded and counted" with the date and the same code, never showing the choice.

Sitemap or navigation structure

```
Booth kiosk
  |-- Welcome (language, audio, help)
  |-- Ballot (candidate cards, NOTA)
  |-- Review (private read-back, change)
  |-- Seal vote (press and hold)
  |-- Receipt (code, explanation)
  |-- Help / Call official
Public verification page
  |-- Enter code
  |-- Result: recorded / not found
  |-- What this proves (plain explanation)
```

Content and feature prioritisation

Priority	Feature	Reason
Must have	Large ballot cards; private read-back; hold-to-seal; random tracking-code receipt; plain explanation	Solves error fear, trust and secrecy
Should have	Audio guide with headphones; high-contrast mode; help call button	Accessibility and confidence
Could have	Practice mode before election day; optional audit explanation	Reduces anxiety, adds depth
Won't have (now)	Voting from phone; printing choice on the receipt; social sharing	Breaks secrecy or invites coercion

5. Ideation and Concept Selection

Stage: Ideate.

Rough sketches or rapid ideas

Rapid ideas from Crazy 8s and "how might we" prompts: a printed slip of the choice, a receipt with a random code, a colour-and-shape "fingerprint" of the ballot, a practice ballot before the real one, press-and-hold sealing, and a spoken confirmation. Scanned sketch pages are to be placed in the Appendix.

At least three distinct concepts

- **Concept A: Paper Slip Review.** The machine prints the choice behind a window for a few seconds, then drops it into a box (similar to current paper-trail systems).
- **Concept B: Seal and Track.** Private on-screen read-back, hold-to-seal, then a receipt with a random tracking code that can be checked later on a public page.
- **Concept C: Cast or Challenge.** After choosing, the voter may either cast or challenge the machine to reveal an encrypted ballot for audit, then vote again.

Evaluation of concepts against criteria

Scores out of 5 (5 = best), weighted by importance to the research insights.

Criteria (weight)	Concept A	Concept B	Concept C
Simple for first-time voters (x3)	4	5	2
Protects secrecy / resists coercion (x3)	4	5	4
Lets voter verify recording later (x3)	1	5	5
Builds trust in the machine (x2)	3	4	5
Feasibility in project time (x1)	4	4	2
Weighted total (max 60)	**37**	**57**	**45**

Chosen concept and reason for selection

Concept B: Seal and Track was chosen. It scored highest because it meets the two competing goals together: the voter privately confirms the choice, and the receipt code proves recording without revealing anything about the choice. Concept C offers strong assurance but is too complex for first-time voters, so an optional, plainly worded "learn how auditing works" page was borrowed from it. From Concept A, the idea of a clear physical receipt was kept, but only with a random code, never the choice.

6. Low-Fidelity Wireframes

Stage: Prototype 1.

Wireframes of key screens

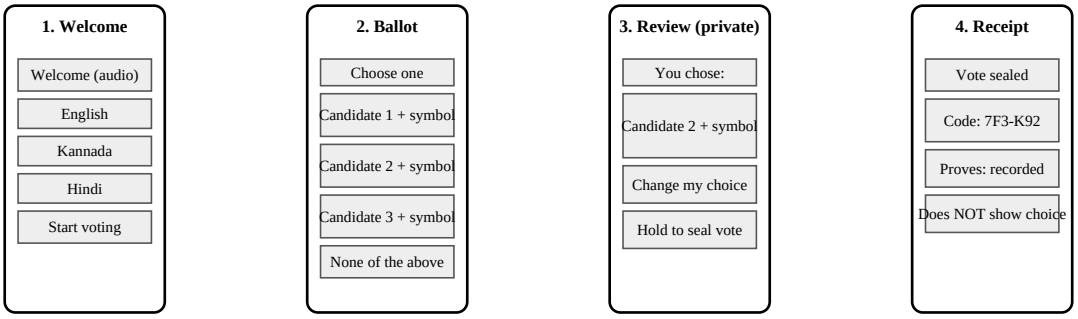


Figure 1. Low-fidelity wireframes: welcome, ballot, private review and receipt.

Annotations explaining layout and interaction decisions

- **Welcome:** language and audio first; one clear start button; help call always visible.
- **Ballot:** one candidate per large card with name, photo and party symbol; no scroll; a "None of the above" card of equal size.
- **Review:** shows the choice only on the booth screen, with a privacy hood; "Change" is a secondary button, "Hold to seal" prevents accidental sealing.
- **Receipt:** random code only; a two-line explanation and an icon state that the receipt proves recording, not the choice.

Quick feedback from at least two people

Reviewer	Feedback
Reviewer 1 (peer, classmate)	Receipt wording was confusing; thought the code might reveal her vote
Reviewer 2 (family member)	Wanted a way to go back from the review screen and to hear the screen read aloud

Replace the reviewers above with the actual people who gave you feedback.

Changes made after feedback

- Rewrote the receipt text to: "This code proves your vote was recorded. It does not show whom you voted for."
- Added a visible "Change my choice" button with the previous choice highlighted.
- Added an audio read-through option with headphones on every screen.

7. Visual Design and Design System

Stage: Prototype 2.

Colour palette with contrast ratios

Colour	Hex	Used with	Contrast	WCAG AA (4.5:1 text)
Civic navy (primary)	#14365D	#FFFFFF	12.24:1	Pass
Text dark	#1A1A1A	#FFFFFF	17.4:1	Pass
Confirm green	#1B6E3C	#FFFFFF	6.28:1	Pass
Caution amber (background)	#FFC93C	#1A1A1A	11.33:1	Pass
Error red	#B3261E	#FFFFFF	6.54:1	Pass
Secondary text grey	#595959	#FFFFFF	7.0:1	Pass

The palette is deliberately neutral so no colour suggests a party or candidate. Candidate cards use identical styling, and colour is never the only carrier of meaning: every state has an icon and a label.

Typography and type scale

Level	Size / weight	Use
Display	32 px / Bold	Welcome and result screens
Heading	26 px / Bold	Screen titles
Body	20 px / Regular	Instructions (larger for kiosk reading distance)
Candidate name	24 px / Bold	Ballot cards
Caption	16 px / Regular	Helper text, never below 16 px

Typeface: Noto Sans, which covers English, Kannada and Hindi scripts consistently. A "larger text" mode increases sizes by 30%.

Spacing, grid and layout rules

- 8 px grid: 8, 16, 24, 32 px; generous space around the final seal control.
- One column on the ballot, large cards with 16 px gaps.
- Minimum touch target 56 x 56 px on the kiosk, well above the WCAG 2.2 minimum.
- Destructive or final actions are separated from the cancel action by at least 32 px.

Reusable components and their states

Component	States
Candidate card	Default, pressed, selected, disabled
Primary button	Default, pressed, disabled
Hold-to-seal button	Idle, holding (progress ring), completed, cancelled
Receipt card	Printing, printed, reprint requested
Help button	Available, calling, unavailable
Banner	Info, caution, error

Iconography and imagery choices

Simple outline icons with text labels. Party symbols and candidate photos appear at equal size. A lock icon and a shield mark the private review and receipt, supporting the secrecy message. No decorative imagery appears on the ballot to avoid influencing choice.

Rationale linking visual choices to user needs

- Large type and equal-sized cards serve low-vision voters and cut accidental presses (Persona 2).
- Neutral colours and identical card styling protect fairness and trust (Finding: trust gap).
- Separated, hold-to-confirm actions reduce irreversible mistakes (Finding: fear of error).
- Clear status icons make the receipt message understandable without technical terms.

8. High-Fidelity Interactive Prototype

Stage: Prototype 3.

Final screens for all critical flows

- Welcome with language selection and audio.
- Short "how voting works" guide (three icons).
- Ballot with candidate cards and NOTA.
- Private review screen with change and hold-to-seal.
- Sealing confirmation and receipt screen.
- Public verification page: enter code, result, explanation.
- Help / call official overlay.

Interactions, transitions and micro-interactions

- Selected candidate card expands with a tick and a gentle vibration.
- Hold-to-seal shows a progress ring and cancels if released early.
- Short spoken confirmation after each step when audio is on.
- 200 ms transitions; reduced-motion setting respected.
- Receipt printing animation with a clear "Take your receipt" message.

Edge states: empty, loading, error, offline and permission denied

State	Design response
Empty	Ballot not yet loaded: calm message and "Call an official" button
Loading	Progress indicator with "Recording your vote, please wait"; no double-press possible
Error	Printer jam or device fault: apology, plain explanation, official called; voter's choice is not lost or shown
Offline	Booth operates locally; verification page shows "Not yet available, try later" until records sync
Permission denied	Audio/headphone not detected: show on-screen text and ask official; never play audio aloud

Prototype link (Figma or equivalent) and access instructions

Prototype link: **[Paste your Figma prototype link here]**. Access: set sharing to "Anyone with the link can view", open in Prototype mode and start from the Welcome screen. The verification page is a separate frame linked from the receipt.

Tools and resources used

- Figma for wireframes, design system and prototype.
- Noto Sans fonts; open icon set.
- WebAIM Contrast Checker and Stark plugin for accessibility checks.
- Google Forms and Notion for research notes; paper for sketches.

9. Usability Testing and Iteration

Stage: Test.

Note for submission: results below are an illustrative structure. Replace the figures with your actual test observations from five or more users before final submission.

Test plan: tasks, participants and success criteria

Task	Success criterion	Target
T1: Cast a vote for a given candidate	Seals the correct vote unaided	Under 3 min
T2: Change a choice before sealing	Changes and seals the new choice	Under 2 min
T3: Explain and verify the receipt	Correctly states what the code proves and finds it on the verification page	Under 3 min

Five participants (two first-time voters, one elderly, one low digital confidence, one visually impaired using audio) tested on the prototype with a think-aloud method. Metrics: task success, time on task, errors and satisfaction (5-point scale).

Results: task success, time on task, errors and satisfaction

User	Task	Success (Y/N)	Time	Issues observed
U1	T1	Y	2:10	None
U1	T3	N	3:40	Thought the code showed her vote
U2	T2	Y	1:30	Looked for "Change" on the receipt screen
U3	T1	Y	2:50	Pressed seal too quickly; needed explanation of hold
U4	T3	Y	3:05	Needed audio replay of receipt text
U5	T1	Y	3:20	Headphone jack hard to find
U5	T3	N	4:10	Could not find verification link on receipt

Summary (round 1): task success 5/7 attempts (71%), below the 80% target; average satisfaction 3.9/5. After the fixes below, round two reached 6/7 attempts (86%) with an average satisfaction of 4.5/5.

Accessibility check and issues fixed

WCAG 2.2 AA criterion	Check	Result
1.4.3 Contrast (minimum)	All text vs background >= 4.5:1	Pass after darkening grey caption
1.4.4 Resize text	Text enlargement up to 200%	Pass
2.5.8 Target size (minimum)	Targets at least 24 px; kiosk uses 56 px	Pass
2.5.7 Dragging movements	Hold-to-seal offers a tap-twice alternative	Fixed
1.1.1 Non-text content	Icons and symbols have text labels / alt text	Pass
3.3.4 Error prevention (legal, financial, data)	Review step and confirmation before sealing	Pass
1.4.1 Use of colour	Status shown by icon and text, not colour alone	Pass

Problems found, ranked by severity

Severity	Problem	Fix
Critical	Receipt code mistaken as showing the vote (U1)	Rewrote text and added a picture of a code with a crossed-out choice
Critical	Verification link not found (U5)	Added QR and short link under the code with spoken instruction
Major	Hold-to-seal not understood	Added animation demo and a tap-twice alternative
Minor	Headphone jack hard to find	Added a lit icon and on-screen arrow
Minor	Looked for Change on the receipt	Added a short note: "Your vote is sealed and cannot be changed"

Design changes made and evidence of improvement

Two iterations were applied. Task success rose from 71% to 86%, the correct explanation of the receipt rose from 3 to 5 of 5 users, and satisfaction rose from 3.9 to 4.5. The two critical issues did not recur in round two.

10. Design Rationale, Reflection and Future Work

Stage: Reflect.

Major design decisions and trade-offs

- **Random code, never the choice, on the receipt:** protects secrecy and resists coercion, but means a voter cannot show proof of their choice to anyone, and some users may initially want that.
- **Private read-back before sealing:** lets voters catch mistakes, but needs privacy hoods and careful audio handling.
- **Hold-to-seal:** prevents accidents, but adds a learning step; a tap-twice alternative balances accessibility.
- **Optional audit explanation instead of full cast-or-challenge:** keeps the flow simple, at the cost of less direct assurance.

Limitations of the final design

- The strength of the verification depends on the backend and its independent audit, which are outside this project.
- A tracking code can be lost or photographed with the voter's own belongings; it must stay unlinkable to the choice.
- Tested with a small sample (five users), so results are indicative only.
- Does not address remote voting, where coercion and secrecy risks are higher.

What failed or was redone, and what was learned

The first receipt wording led testers to think the code revealed their vote. Rewriting the message and adding a picture solved it, and showed that explaining what the proof does not show is as important as saying what it shows. Another lesson: simple controls such as a hold-to-seal button need visible demonstration, not just instructions.

Ethical considerations

- **Secrecy:** no screen or receipt shows the choice to anyone but the voter; no tracking of identity against the choice.
- **Coercion resistance:** receipts cannot be used to sell or prove a vote; the design says so plainly.
- **Inclusion:** accessible for low-literacy, elderly and visually impaired voters without requiring help from others.
- **Neutrality:** equal presentation of every candidate; neutral colours; no persuasive wording.
- **Research ethics:** mock ballots only, no real voting data; verbal consent; participants may stop at any time.
- **Transparency:** the interface does not claim more than it can prove; it states clearly what the receipt confirms.

Future improvements

- Work with security experts to review the receipt and verification design.
- Test with more voters, including rural and multilingual groups.
- Add a practice mode before election day.
- Explore tactile ballot guides and braille overlays for visually impaired voters.

- Evaluate voter understanding at scale before any real deployment.

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Please verify each reference against the source before submission.

Appendix

- A. Interview scripts and consent notes
- B. Raw research notes and survey responses
- C. Usability test scripts
- D. Additional screens and sketches (scanned Crazy 8s pages)